
FINAL PROJECT

STA 5384/4384

March 24, 2026

Please follow the following instructions for your final project.

1. Find a publicly available multivariate dataset, that we did not use during the course **and that you did not use on assignments 2, 3, 4, and 5**, that would be appropriate for a detailed multivariate analysis. Using a dataset that is not publicly available, a dataset we used in the course, or that you used on your assignments will result in a grade of zero.
2. Perform an exploratory analysis on your data (i.e. visualizations, means, covariances, etc.)
3. Perform a detailed analysis using three of the following four topic areas we considered during the course (PCA, Cluster Analysis, LDA/QDA, Bagging/Random Forests).
4. Write up your findings in a report according to the parameters below.

Notes on the Report:

- Plagiarism (including the use of AI to write the report) will not be tolerated and will result in a grade of zero.
- A hard copy report not exceeding 10 pages (including all tables and figures, but not a title page) is due for all students regardless of presentation date by **2pm on April 28, 2026**. The font size must be 11pt font with 1.5 line spacing. Latex is strongly encouraged, but not necessary.

- This report should **NOT** include R code or raw R output. I.e. you should summarize all R output via formatted tables and figures.
- R code must be submitted to me at the email address Michael.Gallaugh@baylor.edu by **2pm on April 28, 2026**. R code must be submitted in a self contained .R file that includes **ALL** your analyses. Furthermore, this R code should be commented, but do not include interpretations in the comments. You should ensure that the R file runs without errors before submission. If your dataset is not an R dataset, also include your dataset, and in your code, include code that loads the dataset.

Notes on the Oral Presentations:

- Oral presentations will take place on **April 28, April 30, and May 5, 2026 during class**. You will be randomly assigned a day about one to two weeks before the presentations. You only need to attend class on the day that you present. If you do not present on the first day, either submit your report to me in my office by the due date/time, or show up at the beginning of class on the first day of presentations to hand it in.
- If for some reason you cannot make your presentation date (due to a reasonable conflict) let me know **immediately** after the schedule is posted.
- You will have around 8 minutes to present a quick summary of your data, and the analyses you performed. This will be followed by a brief discussion period.
- **Before 12pm on April 27, 2026**, you must email slides for your presentation in **pdf format**. After this deadline no significant alterations can be made to your slides to be fair to students presenting on earlier days.
- Your presentation grade will be based on the accuracy and clarity of your presentation and adherence to the time limit.
- Due to the time constraint, there will be a hard cut off at 8 minutes for your presentation.

(Tentative) Grading Criteria:

- R coding and implementation (10 marks)
- Interpretation of results (10 marks)
- Visualization and analysis results (10 marks)
- Understanding of concepts (10 marks)
- Clarity and presentation of the report (5 marks)
- Oral Presentations (5 marks)

Good luck with the final project! If you have any questions or need clarification, feel free to reach out to me. Happy analyzing!